Getting started with sloshR

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sloshR turns SLOSH storm runs (rex files) into tables and figures in R. This vignette uses the two **synthetic** example storms shipped with the package; with real data, point `txt_dir` at your converted folder instead.

1. Convert rex files (Windows, once)

Rex files are in NOAA's own binary format. `slosh_convert()` runs NOAA's SLOSH Display Program library in batch and writes every frame as text:

```
log <- slosh_convert("C:/slosh.pkg/sloshdsp")   # all rex files
log[status != "ok"]                              # any failures
txt_dir <- "C:/slosh.pkg/sloshdsp/output/rex_txt"
```

Everything below works on the converted text on any platform.

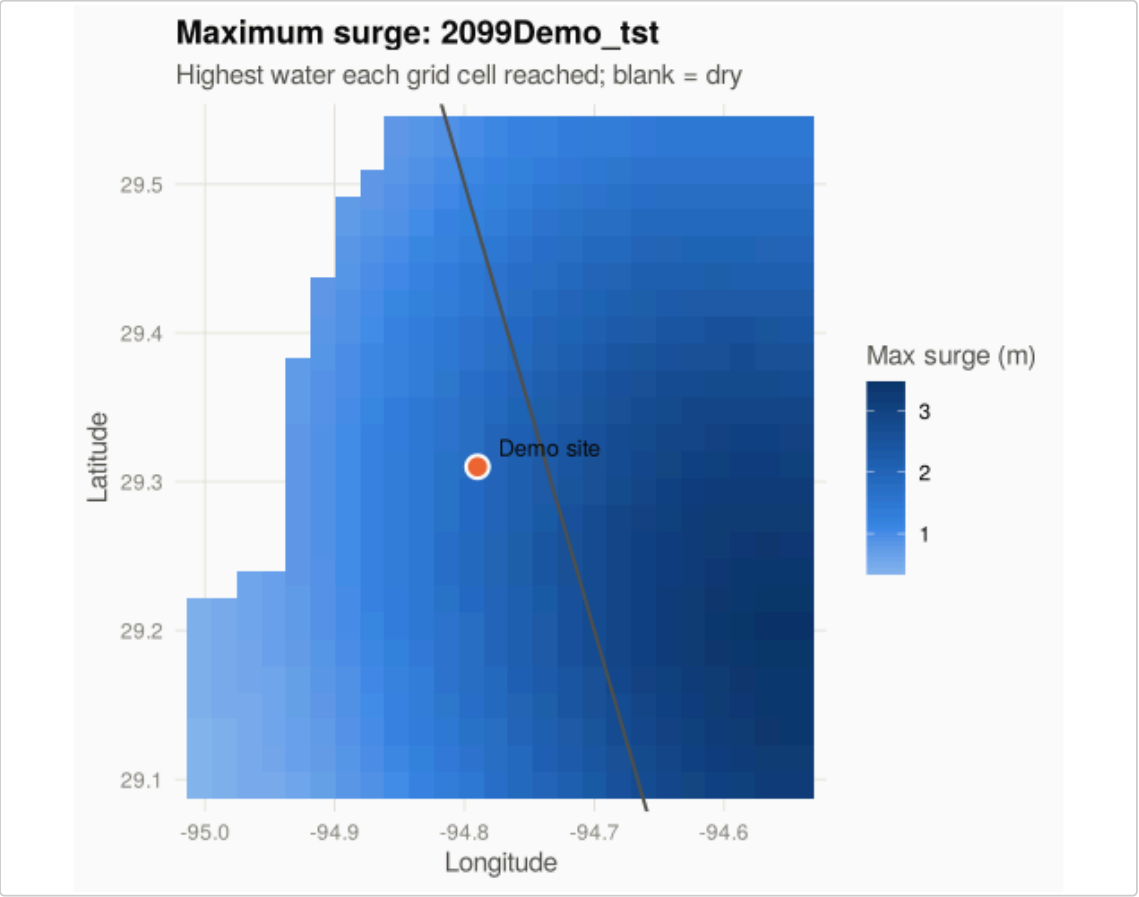
```
txt_dir <- slosh_example_dir()
slosh_list_storms(txt_dir)
#>             storm year   name basin n_frames
#> 1: 2098Sample_tst 2098 Sample   tst       18
#> 2:   2099Demo_tst 2099   Demo   tst       18
```

2. Read one storm

```
fr  <- slosh_read_frames("2099Demo_tst", txt_dir)   # every cell, every frame
tr  <- slosh_read_track("2099Demo_tst", txt_dir)    # storm centre and winds
env <- slosh_envelope(fr)                            # max surge per cell
head(env)
#>           storm cell lon    lat max_surge_ft max_surge_m time_of_max
#> 1: 2099Demo_tst    1 -95 29.100          1.1     0.33528 2099-09-10 09:00:00
#> 2: 2099Demo_tst    2 -95 29.118          1.2     0.36576 2099-09-10 09:30:00
#> 3: 2099Demo_tst    3 -95 29.136          1.2     0.36576 2099-09-10 09:30:00
#> 4: 2099Demo_tst    4 -95 29.154          1.3     0.39624 2099-09-10 09:30:00
#> 5: 2099Demo_tst    5 -95 29.172          1.3     0.39624 2099-09-10 09:30:00
#> 6: 2099Demo_tst    6 -95 29.190          1.4     0.42672 2099-09-10 10:00:00
#>     cell_km ds_deg
```

```
#> 1: 2.001509  0.018
#> 2: 2.001509  0.018
#> 3: 2.001509  0.018
#> 4: 2.001509  0.018
#> 5: 2.001509  0.018
#> 6: 2.001509  0.018
```

```
site <- list(name = "Demo site", lat = 29.31, lon = -94.79)
slosh_plot_map(env, tr, site)
```

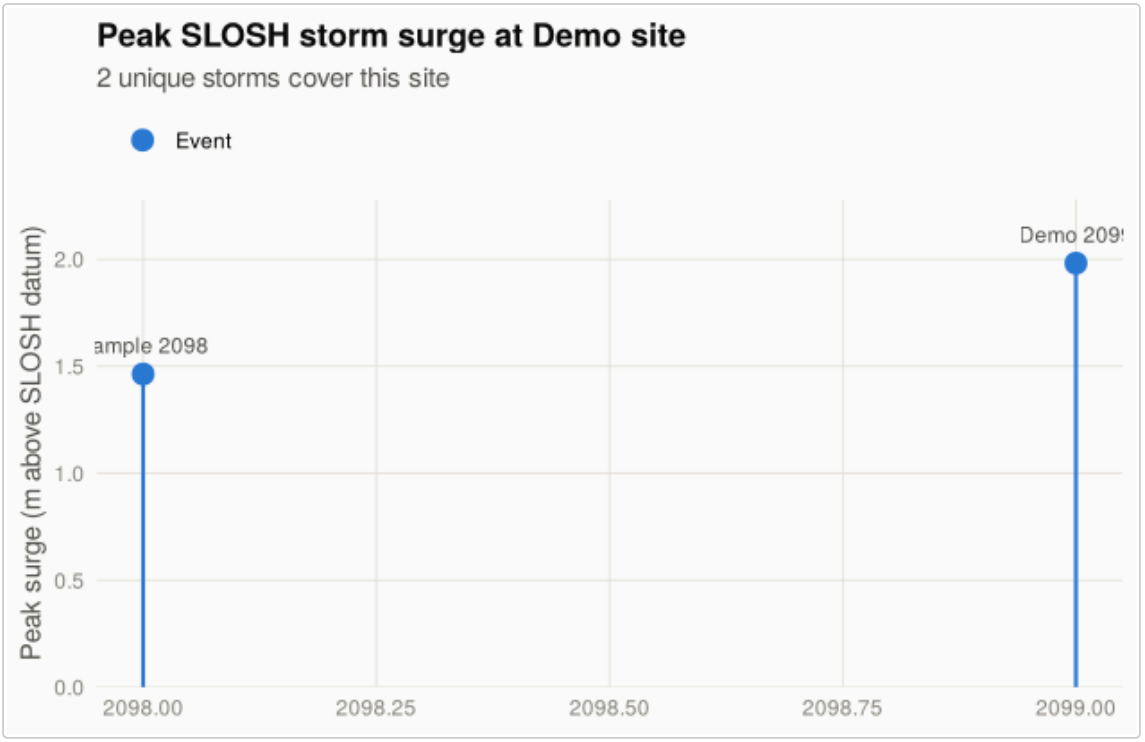


3. The event table at a site

`slosh_events()` reads every storm and returns one row per storm with its peak surge at the site. Storms whose grid does not reach the site have `covers_site = FALSE`; repeat runs of the same storm are flagged in `duplicate_event`.

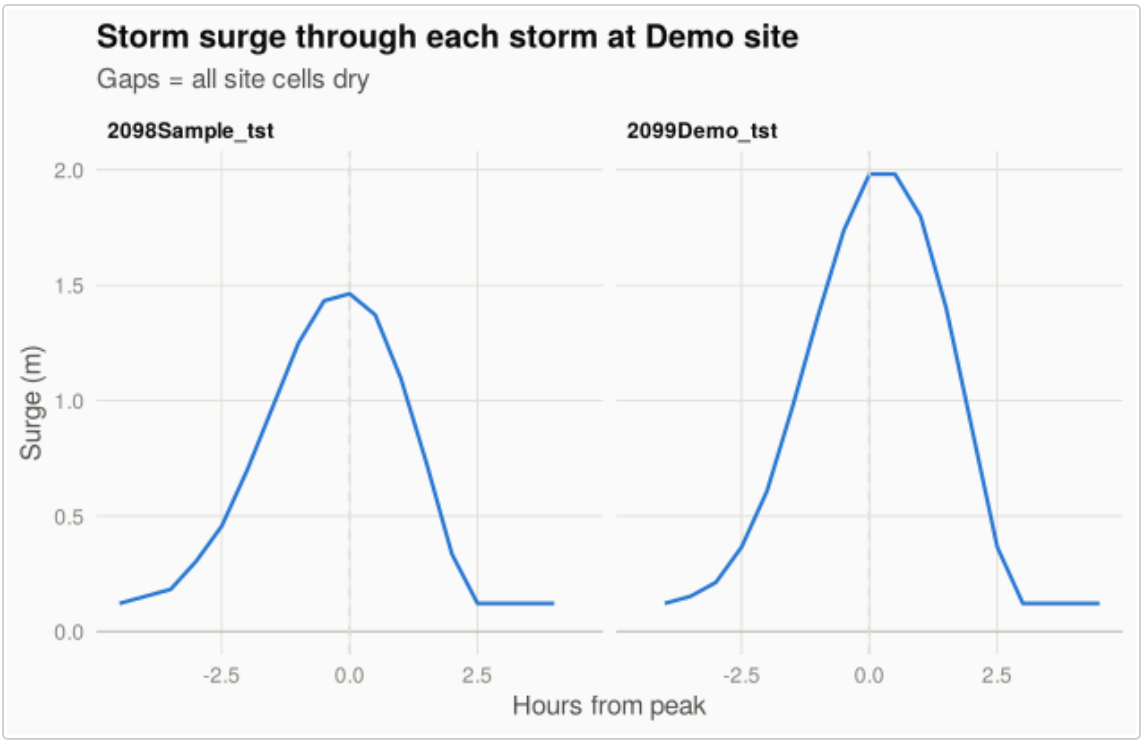
```
ev <- slosh_events(txt_dir, lat = site$lat, lon = site$lon, radius_km = 3)
#>   2098Sample_tst
#>   2099Demo_tst
#> 2 storms; 2 cover the site; 2 unique events at the site.
ev[, .(storm, covers_site, peak_surge_m, peak_time_utc, duplicate_event)]
#>             storm covers_site peak_surge_m       peak_time_utc
duplicate_event
#> 1: 2098Sample_tst        TRUE        1.463 2098-08-20 04:30:00
```

```
FALSE
#> 2:   2099Demo_tst        TRUE        1.981 2099-09-10 10:00:00
FALSE
slosh_plot_peaks(ev, site$name)
```



4. Storm metrics and flood footprint

```
ts <- rbind(
  slosh_site_series(slosh_read_frames("2098Sample_tst", txt_dir),
                    site$lat, site$lon),
  slosh_site_series(fr, site$lat, site$lon))
slosh_storm_metrics(ts, thresholds = c(0.5, 1, 2))
#>             storm  peak_m           peak_time step_h hours_wet
max_rise_m_per_h
#> 1: 2098Sample_tst 1.46304 2098-08-20 04:30:00    0.5         9
0.54864
#> 2:   2099Demo_tst 1.98120 2099-09-10 10:00:00    0.5         9
0.79248
#>    rise_h hours_above_0.5m hours_above_1m hours_above_2m
#> 1:    4.5              4.0            2.5              0
#> 2:    4.0              4.5            3.0              0
slosh_plot_series(ts, site$name)
```




```
slosh_footprint(env, thresholds = c(1, 2, 3))
#>           storm threshold_m n_cells area_km2
#> 1: 2099Demo_tst           1     443   1774.7
#> 2: 2099Demo_tst           2     236    945.4
#> 3: 2099Demo_tst           3      80    320.5
```

5. Compound timing

Did the surge peak coincide with high tide, or with peak rainfall or river flow? Supply the other series as a data frame with `time` and a value.

```
t0 <- min(ts$time) - 86400
tide <- data.frame(time = t0 + seq(0, 3 * 86400, by = 600))
tide$level <- 0.4 * cos(2 * pi * as.numeric(tide$time - t0) / (12.42 *
3600))
slosh_tide_timing(ts, tide)
#>             storm           peak_time  peak_m tide_at_peak_m
total_at_peak_m
#> 1: 2098Sample_tst 2098-08-20 04:30:00 1.46304     -0.1108384
1.352202
#> 2:   2099Demo_tst 2099-09-10 10:00:00 1.98120             NA
NA
#>      nearest_high_tide lag_to_high_tide_h max_tide_12h_m
#> 1: 2098-08-20 00:50:00           3.666667      0.3999977
#> 2:                <NA>                 NA             NA

pk <- ts[storm == "2099Demo_tst"][which.max(surge_m), time]
rain <- data.frame(time = pk + seq(-48, 48) * 3600)
rain$value <- 40 * exp(-((seq(-48, 48) - 10) / 6)^2)
```

```
slosh_driver_lag(ts[storm == "2099Demo_tst"], rain)
#>           storm     surge_peak_time surge_peak_m    driver_peak_time
#> 1: 2099Demo_tst 2099-09-10 10:00:00       1.9812 2099-09-10 20:00:00
#>    driver_peak lag_h driver_at_surge_peak
#> 1:          40    10             2.487061
```

6. Validation and transfer to ungauged locations

```
obs <- data.frame(storm = ev$storm, obs_peak_m = ev$peak_surge_m + c(0.2,
-0.1))
slosh_validate(ev, obs)
#> SLOSH vs observed peaks: 2 storms
#>   bias -0.05 m | MAE 0.15 m | RMSE 0.16 m | r NA
#>             storm peak_surge_m       peak_time_utc obs_peak_m error_m
#> 1: 2098Sample_tst        1.463 2098-08-20 04:30:00      1.663    -0.2
#> 2:   2099Demo_tst        1.981 2099-09-10 10:00:00      1.881     0.1

targets <- data.frame(target = c("Bay head", "Open coast"),
                      lat = c(29.36, 29.26), lon = c(-94.86, -94.74))
slosh_transfer(txt_dir, gauge = list(lat = 29.31, lon = -94.79), targets)
#> Target-to-gauge peak surge ratios
#>        target n_storms median_ratio       q25       q75 min_ratio
max_ratio
#> 1:   Bay head        2    0.6668269 0.6564904 0.6771635 0.6461538
0.687500
#> 2: Open coast        2    1.2684295 1.2487981 1.2880609 1.2291667
1.307692
```

A stable target-to-gauge ratio across storms lets a long gauge record's frequency analysis be carried to the targets.

7. Animation

```
slosh_animate("2099Demo_tst", txt_dir, file = "demo.gif", site = site)
```

Appropriate use

SLOSH is built for hurricane planning; NOAA advises against using its output for engineering design. The NOAA rex library holds only a few storms per basin, so frequency or trend analysis needs a long observed record, with SLOSH used for footprints, timing, validation and transfer.